

Fashionpedia

Attribute Extraction System

End-to-End Fashion Attribute Recognition · FastAPI Deployment · LLM-Powered Descriptions

156,496

Training Crops

101

Attribute Classes

0.3619

Validation mAP

3-Stage

Inference Pipeline

Problem Statement

Scale

45,623 images × 294 possible attributes — manual tagging is slow and inconsistent

Multi-Label

Each garment has multiple simultaneous attributes (avg 3.88 per crop at inference)

Extreme Imbalance

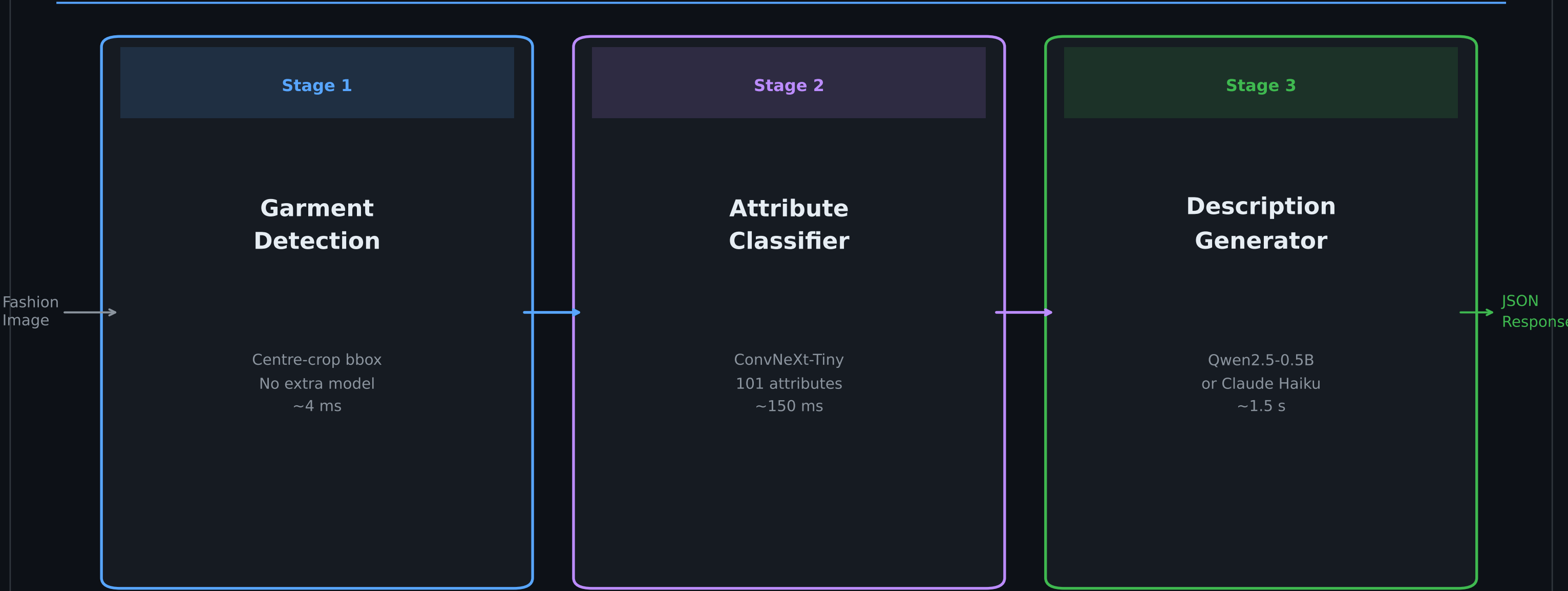
Rarest attribute (paisley) is 515× more rare than the most common class

Annotation Noise

38% of raw annotations have zero attributes — unannotated, not confirmed negatives

Goal: Train a model that takes any fashion image and outputs structured, confidence-scored attribute labels — automatically.

Proposed Solution: 3-Stage Inference Pipeline



Dataset & Training Pipeline

Fashionpedia 2020

45,623 images · 333,401 raw annotations

Drop Zero-Attr

−38.1% (unannotated instances discarded)

Drop Small Boxes

−7.8% (bbox area < 1,024 px² or side < 16 px)

Dedup Check

−0.02% (38 duplicate crops removed)

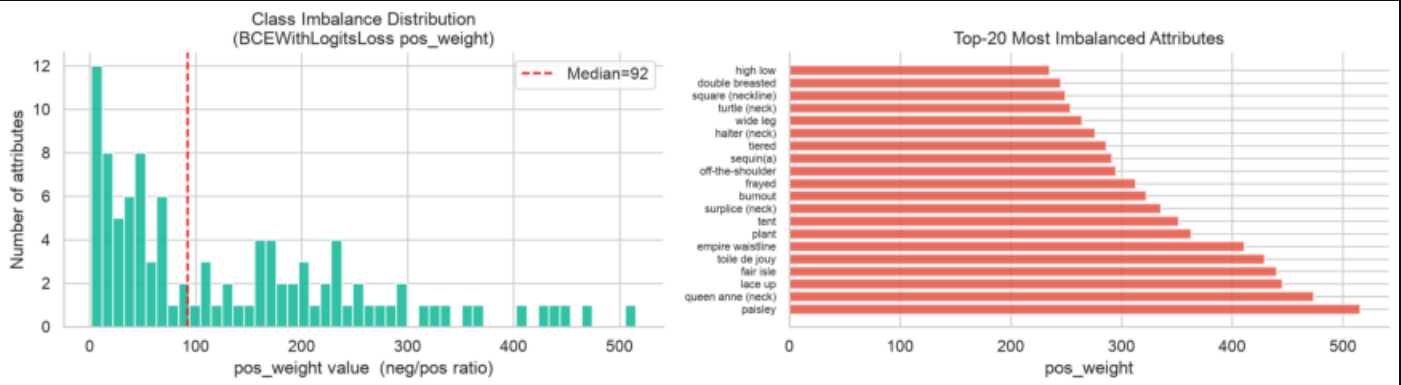
Filter Rare Attrs

294 → 101 classes (≥ 300 training samples per class)

Final Split

156,496 train crops · 4,066 val crops · 101 classes

Class Imbalance: pos_weight per Attribute
(median 92.5×, max 515.5× for 'paisley')



8 Supercategories retained:
length

neckline type

silhouette

textile pattern

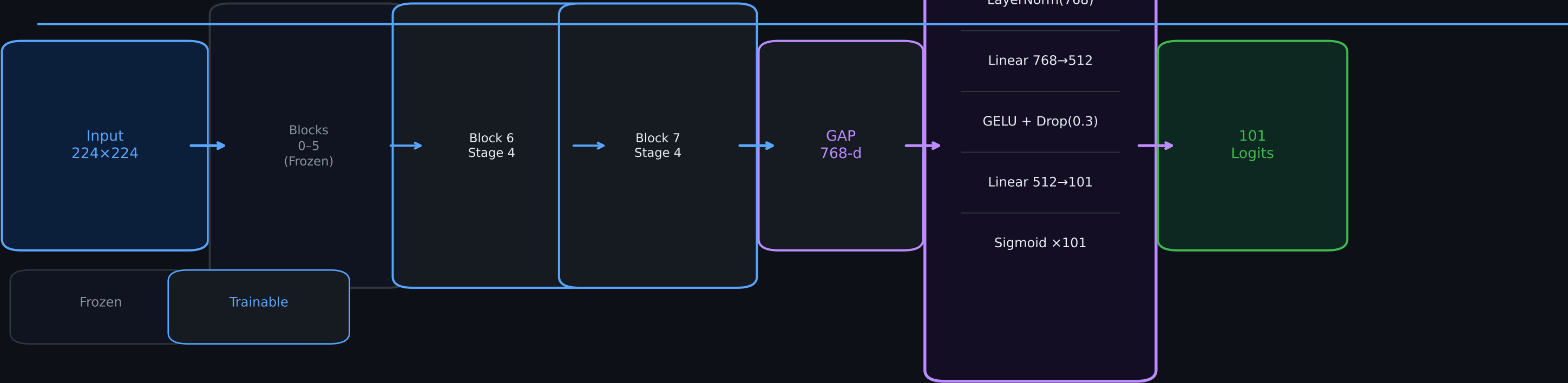
waistline

opening type

non-textile material

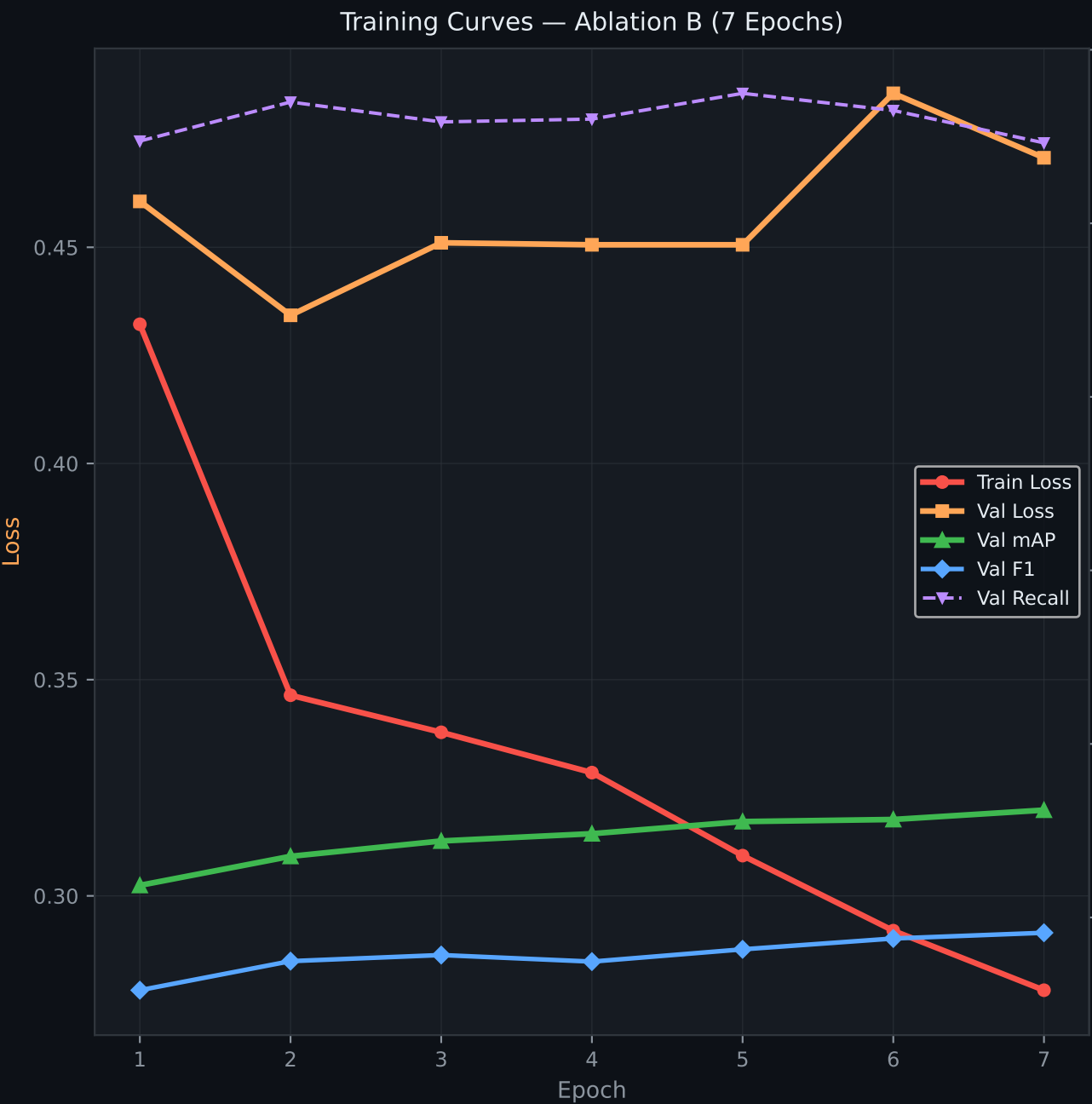
mfg techniques

Model Architecture: ConvNeXt-Tiny + Partial Fine-Tuning



Partial Fine-Tuning Ablations		
Config	Trainable Params	Result
A Head Only	~0.5M (1.8%)	Baseline
B Stage-4 + Head ✓	~1.2M (3.2%)	SELECTED
C Stages 3-4 + Head	~2.5M (8.9%)	Extended
LR head=1e-3 · LR backbone=1e-4 (10:1 ratio) · 3-epoch warmup · Cosine Annealing		

Results & Example Predictions



Best Checkpoint (Epoch 7)

Val mAP

0.3619

F1 (macro)

0.2912

Recall

0.7464

Precision

0.2023

Live Inference — any_dress_photo.jpg
threshold=0.45 · 11 attributes detected · ~75 ms

plain (pattern)		99%
no non-textile material		97%
no waistline		96%
symmetrical		95%
hip (length)		94%
no special manufacturing technique		93%
regular (fit)		92%
loose (fit)		70%
above-the-hip (length)		69%

Deployment Architecture: FastAPI + Multi-Backend LLM

FastAPI Application



Total Latency: ~4 ms detection + ~150 ms attributes + ~1.5 s LLM = ~1.7 s (local Qwen) / ~0.9 s (Claude Haiku API)

Learning Experience & Future Work

Model Architecture Journey



Key Technical Decisions

BCEWithLogitsLoss + pos_weight

Multi-label binary CE with per-class weighting handles 515× imbalance. Weight cap=50 prevents gradient explosion on rarest classes.

Partial Fine-Tuning (Stage 4 + Head)

Freezing 6 of 8 blocks with only 1.2M trainable params prevented overfitting and kept each epoch under 26 minutes on M-series MPS.

Differential Learning Rates + Warmup

10:1 ratio (1e-3 head / 1e-4 backbone) + 3-epoch linear warmup protects pre-trained ImageNet representations while the new head learns fast.

Annotation-Level Crops as Training Units

Feeding in 100× crops (not full images) aligns inputs with attribute labels — each crop = one garment region described by its annotation.

Future Work: More training epochs · iOS SwiftUI app · YOLOv8 garment detector · Saliency-based cropping · Larger ConvNeXt variant